

CSL7120: DL OPs Assignment-1 Report

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Question - 1 (Part-A)

Introduction: Image Classification on MNIST and FashionMNIST Datasets, using Resnet18 and Resnet50 Models, with varying hyperparameter values, like batch_size, learning rate, epochs, pin_memory and optimizers.

Hyperparameter values:

- **Batch_size :** 16, 32
- **Epochs:** 1, 2
- **Learning rates:** 0.001, 0.0001
- **Optimizers:** SGD, Adam
- **Pin_memory:** True, False

Results:

For MNIST dataset (epochs = 1, pin_memory=True)
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Batch size	Optimizer	Learning Rate	Test Classification Accuracy (in %)	
			Resnet18	Resnet50
16	SGD	0.001	97.64	97.81
16	SGD	0.0001	86.97	59.47

16	Adam	0.001	98.08	97.62
16	Adam	0.0001	98.52	98.82
32	SGD	0.001	97.05	96.57
32	Adam	0.0001	64.83	50.24
32	SGD	0.001	98.06	96.85
32	Adam	0.0001	97.62	98.16

For MNIST dataset (epochs = 1, pin_memory=False)

			Test Classification Accuracy (in %)	
Batch size	Optimizer	Learning Rate	Resnet18	Resnet50
16	SGD	0.001	98.06	97.66
16	SGD	0.0001	91.33	68.24
16	Adam	0.001	98.32	97.88
16	Adam	0.0001	98.57	98.38
32	SGD	0.001	96.84	96.83
32	Adam	0.0001	71.05	43.84
32	SGD	0.001	98.53	97.80
32	Adam	0.0001	98.45	97.83

For MNIST dataset (epochs = 2, pin_memory=True)

			Test Classification Accuracy (in %)	
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Batch size	Optimizer	Learning Rate	Resnet18	Resnet50
16	SGD	0.001	98.70	98.41
16	SGD	0.0001	95.29	89.75
16	Adam	0.001	98.77	98.07
16	Adam	0.0001	99.17 (Best)	98.40
32	SGD	0.001	97.65	97.88
32	Adam	0.0001	82.75	64.76
32	SGD	0.001	98.30	97.78
32	Adam	0.0001	98.92	97.76

For MNIST dataset (epochs = 2, pin_memory=False)

			Test Classification Accuracy (in %)	
Batch size	Optimizer	Learning Rate	Resnet18	Resnet50
16	SGD	0.001	99.04	98.30
16	SGD	0.0001	96.66	88.26
16	Adam	0.001	98.78	98.53
16	Adam	0.0001	99.07	98.79
32	SGD	0.001	98.48	97.94
32	Adam	0.0001	90.62	68.72
32	SGD	0.001	98.98	97.71
32	Adam	0.0001	98.85	97.93

For FashionMNIST dataset (epochs = 1, pin_memory=True)

			Test Classification Accuracy (in %)	
Batch size	Optimizer	Learning Rate	Resnet18	Resnet50
16	SGD	0.001	83.69	83.98
16	SGD	0.0001	73.22	66.59
16	Adam	0.001	84.33	84.26
16	Adam	0.0001	89.53	86.18
32	SGD	0.001	74.46	71.20
32	Adam	0.0001	71.08	58.66
32	SGD	0.001	87.50	68.04
32	Adam	0.0001	84.48	85.50

For FashionMNIST dataset (epochs = 1, pin_memory=False)

			Test Classification Accuracy (in %)	
Batch size	Optimizer	Learning Rate	Resnet18	Resnet50
16	SGD	0.001	85.96	80.10
16	SGD	0.0001	74.37	65.76
16	Adam	0.001	87.64	85.91
16	Adam	0.0001	88.05	85.48
32	SGD	0.001	83.77	79.65
32	Adam	0.0001	73.34	56.92
32	SGD	0.001	85.68	85.34
32	Adam	0.0001	88.21	84.39

For FashionMNIST dataset (epochs = 2, pin_memory=True)

			Test Classification Accuracy (in %)	
Batch size	Optimizer	Learning Rate	Resnet18	Resnet50
16	SGD	0.001	87.20	86.96
16	SGD	0.0001	79.36	72.13
16	Adam	0.001	88.92	87.51
16	Adam	0.0001	89.98	88.82
32	SGD	0.001	85.58	84.26
32	Adam	0.0001	75.40	66.59
32	SGD	0.001	89.00	87.89
32	Adam	0.0001	89.52	84.44

For FashionMNIST dataset (epochs = 2, pin_memory=False)

			Test Classification Accuracy (in %)	
Batch size	Optimizer	Learning Rate	Resnet18	Resnet50
16	SGD	0.001	88.44	86.85
16	SGD	0.0001	79.64	67.52
16	Adam	0.001	87.28	88.77
16	Adam	0.0001	89.97	89.44
32	SGD	0.001	85.53	84.49
32	Adam	0.0001	75.39	65.26
32	SGD	0.001	88.72	86.30
32	Adam	0.0001	90.28 (Best)	87.97

Question - 1 (Part B)

Introduction: Image Classification on MNIST and FashionMNIST Datasets, using SVM Model, with varying hyperparameter values, like C, gamma, kernel and degrees of polynomial kernel.

Hyperparameter values:

- **C:** 0.1, 1, 10
- **Gamma:** scale, auto
- **Kernels:** rbf, polynomial
- **Degree of polynomial:** 2, 3
- **Max samples:** 20,000

Results:

For MNIST Dataset

C Value	Kernel	Gamma	Test Accuracy	Train Time (in ms)
0.1	rbf	scale	92.95	28369.39
0.1	rbf	auto	86.83	57821.00
1	rbf	scale	96.05	13599.54
1	rbf	auto	91.65	19822.32
10	rbf	scale	96.70 (best)	12867.16
10	rbf	auto	93.33	10187.13
0.1	poly (deg = 2)	scale	92.65	23906.48
0.1	poly (deg = 3)	scale	91.55	28056.12

0.1	poly (deg = 2)	auto	17.68	96021.53
0.1	poly (deg = 3)	auto	11.20	95916.95
1	poly (deg = 2)	scale	95.50	10886.34
1	poly (deg = 3)	scale	95.12	13164.86
1	poly (deg = 2)	auto	81.78	60037.52
1	poly (deg = 3)	auto	18.50	95394.09
10	poly (deg = 2)	scale	96.08	9271.96
10	poly (deg = 3)	scale	95.58	10222.79
10	poly (deg = 2)	auto	92.40	24805.38
10	poly (deg = 3)	auto	72.32	64562.68

For Fashion MNIST Dataset

C Value	Kernel	Gamma	Test Accuracy	Train Time (in ms)
0.1	rbf	scale	80.88	24278.35
0.1	rbf	auto	75.17	37888.52
1	rbf	scale	86.85	13853.05
1	rbf	auto	87.17	18687.21
10	rbf	scale	87.83 (best)	13029.95

10	rbf	auto	86.12	11637.92
0.1	poly (deg = 2)	scale	79.77	21518.59
0.1	poly (deg = 3)	scale	77.62	22872.94
0.1	poly (deg = 2)	auto	55.83	71484.47
0.1	poly (deg = 3)	auto	26.02	92839.85
1	poly (deg = 2)	scale	85.58	12890.00
1	poly (deg = 3)	scale	83.38	14940.31
1	poly (deg = 2)	auto	72.78	35259.57
1	poly (deg = 3)	auto	57.03	56988.61
10	poly (deg = 2)	scale	87.12	11479.55
10	poly (deg = 3)	scale	85.60	12927.32
10	poly (deg = 2)	auto	80.80	19750.59
10	poly (deg = 3)	auto	71.58	32520.86

Analysis:

- Effect of Kernel choice:** The kernel has the strongest impact on SVM performance. RBF consistently outperforms polynomial kernels on both MNIST and Fashion-MNIST because it can model complex, non-linear decision boundaries effectively. Polynomial kernels, especially with higher degrees, struggle on high-dimensional image data and are more sensitive to hyperparameters. Fashion-MNIST, being more complex than MNIST, further highlights RBF's superiority in capturing subtle visual patterns.

- **Effect of C (Regularization):** Increasing C generally improves accuracy up to a point by reducing bias and allowing the model to fit training data more tightly. For both datasets, performance improves from C=0.1 to C=10, with best results mostly at C=10. However, higher C slightly increases overfitting risk and training time. Low C values under-regularize the model, leading to poorer decision boundaries and reduced accuracy.
 - **Effect of Gamma:** Gamma controls the influence of individual samples. The scale setting consistently yields better accuracy and faster convergence than auto, as it adapts gamma to data variance. Auto often leads to overly large or small gamma values, causing underfitting or excessive computation, especially for polynomial kernels. High training times and poor accuracy with auto indicate unstable decision surfaces, particularly on complex datasets like Fashion-MNIST.
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Question - 2

Introduction: Image Classification on **FashionMNIST** dataset, using Resnet18 and Resnet50, By varying hyperparams like optimizer and compute device, and reporting FLOPs involved, test accuracy and Train time.

Hyperparameter values:

- **Optimizer:** SGD, Adam
- **Device:** CPU, GPU
- **Batch size:** 16
- **Learning Rate:** 0.001
- **Epochs:** 1

Results:

				Test Accuracy	Train Time (ms)	GFLOPs
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Device	Batch size	Optimizer	LR	Resnet18	Resnet50	Resnet18	Resnet50	Resnet18	Resnet50
CPU	16	SGD	0.001	86.56	76.81	326988 0.52	103601 76.85	1.824	4.132
CPU	16	Adam	0.001	85.46	81.40	309095 6.7	105343 33.43	1.824	4.132
GPU	16	SGD	0.001	83.38	82.69	121377 .28	239964 .80	1.824	4.132
GPU	16	Adam	0.001	83.81	84.35	127455 .77	250968 .11	1.824	4.132

Notebook Links

- Question 1: [🔗 question-1-dl-ops-assignment-1-b22cs001.ipynb](#)
- Question 2: <https://www.kaggle.com/code/abhaykashyapb22cs001/question-2-dl-ops-assignment-2-b22cs001>
- Best Model Save: [🔗 Best Model for MNIST & Fashion MNIST.ipynb](#)
- Github Repository: [abhaykashyap31/MLOps-AbhayKashyap-B22CS001](#)